Note in Advanced Statistics

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Abstract

The note is about something that I have not learned in my undergraduate course. The courses' main reference [1]. Thanks to my teacher Yaowu Zhang.

The note is organized as follows. In chapter6 Principals of Data Reduction, I introduce some concept about **sufficient statistic, minimal sufficient statistic, ancillary statistics, complete statistics** and some related theorems. In chapter7 point estimation, answer the question why MLE is a good estimator. In chapter8 hypothesis testing, introduce the concept about **power function, UMP and distinguish levle α and size α**. A key theorem here is Neyman-Pearson Theorem.

Keywords: Sufficient Statistic, Factorization Theorem, Basu's Theorem, Complete Statistic,power function,UMP, levle α, size α,Neyman-Pearson Theorem

1 Chatper 6 Principals of Data Reduction

1.1 Sufficient Statistics

There are many examples in slides, like binomial sufficient statistic, sufficient order statistic, uniform sufficient statistic, normal sufficient statistic.

And for the exponential family, there are some remarkable properties.

Let $X_1, \ldots, X_n$ be iid observations from a pdf or pmf that belongs to an exponential family given by

$$f(x \mid \boldsymbol{\theta}) = h(x)c(\boldsymbol{\theta}) \exp\left(\sum_{i=1}^{k} \omega_i(\boldsymbol{\theta}) t_i(x)\right)$$

where $\theta = (\theta_1, \ldots, \theta_d)$, $d \leq k$. Then

$$T(\mathrm{X}) = \left(\sum_{j=1}^{n} t_1\left(X_j\right), \ldots, \sum_{j=1}^{n} t_k\left(X_j\right)\right)$$

that is a sufficient statistic for θ.

Numerous sufficient statistics: If $T(X)$ is a sufficient statistic and satisfies $T(X) = \varphi(S(X))$ then $S(X)$ is also a sufficient statistic.

1.2 Factorization Theorem

$$f(x|\theta) = g(T(x)|\theta)h(x)$$

1.3 Is there exist one sufficient statistic better than any one?——Minimal sufficient statistics

Definition 1.1. If for any other sufficient statistic $T'(X)$, $T(x)$ is a function of $T'(X)$

1.3.1 Way to Find Minimal Sufficient Statistics

Lemma 1.1. *Suppose $f(x)$ and $g(x)$ are two functions and $g(x_1) = g(x_2)$ implies $f(x_1) = f(x_2)$.$\forall x_1, x_2 \in \mathbb{R}$. Then there must exist some function $\varphi(\cdot)$ such that $f(x) = \varphi(g(x))$*

Lemma 1.2. *$T(\mathrm{X})$ is a minimal sufficient statistic if and only if:*

- *$T(X)$ is a sufficient statistic*
- *for any sufficient statistic $T^*(\mathrm{X}), T^*(x_1) = T^*(x_2)$ implies $T(x_1) = T(x_2)$, $\forall x_1, x_2 \in \mathcal{X}$*

1.2 2 means that $T(x)$ is minimal sufficient statistics by $\Leftarrow T(x) = \varphi(T^*(x))$ for any sufficient $T^*(x)$

Theorem 1.3. *Let $f(x|\theta)$ be the pmf or pdf of a sample X. Suppose there exists a function $T(x)$ such that, for every two sample points x and y, the ration $\frac{f(x|\theta)}{f(y|\theta)}$ is a constant as a function of θ if and only if $T(x) = T(y)$. Then $T(x)$ is a minimal sufficient statistic for θ.*

Proof. Sufficient: Let $U = T(x) = x$, so $T^{-1}(T(x)) = y$ when restriction order and module. (Though we get x,y from this condition, but x,y are still arbitrary. So T satisfies the condition every two sample points.)

We have

$$T(x) = T(y)$$

$$f(x|\theta) = f(y|\theta)$$

$$\Rightarrow f(x|\theta) = \frac{f(x|\theta)}{f(y|\theta)} f(y|\theta)$$

The first term is independent to θ, $f(y|\theta) = f(T^{-1}(T(x))|\theta))$. By the Factorization theorem, $T(x)$ is a sufficient statistic.

Minimal:

$f(x) = g(T(x)|\theta)h(x)$ Assume $T^*(x_1) = T^*(x_2)$, $\frac{f(x_1|\theta)}{f(x_2|\theta)} = \frac{g(T^*(x_1)|\theta)h(x_1)}{g(T^*(x_2)|\theta)h(x_2)}$,where $\frac{g(T^*(x_1)|\theta)}{g(T^*(x_2)|\theta)}$ is constant as for $theta$ i.e. is indepedent of x_1 and x_2, $\iff T(x_1) = T(x_2)$

□

Example 1.1. Let $X_1, \ldots, X_n$ be i.i.d $N\left(\mu, \sigma^2\right)$ r.v.s, both μ and σ^2 are unknown

Let x and y denote two sample points, and let $\left(\bar{x}, s_x^2\right)$ and $\left(\bar{y}, s_y^2\right)$ be the sample means and variances corresponding to the x and y samples, respectively. Then,

$$\begin{aligned}\frac{f\left(x \mid \mu, \sigma^2\right)}{f\left(y \mid \mu, \sigma^2\right)} &= \frac{\exp\left[-\left\{n(\bar{x}-\mu)^2+(n-1)s_x^2\right\}/\left(2\sigma^2\right)\right]}{\exp\left[-\left\{n(\bar{y}-\mu)^2+(n-1)s_y^2\right\}/\left(2\sigma^2\right)\right]} \\ &= \exp\left[\left\{-n\left(\bar{x}^2-\bar{y}^2\right)+2n\mu(\bar{x}-\bar{y})\right.\right. \\ &\left.\left.-(n-1)\left(s_x^2-s_y^2\right)\right\}/\left(2\sigma^2\right)\right]\end{aligned}$$

This ratio will be constant as a function of μ and σ^2 if and only if $\bar{x}=\bar{y}$ and $s_x^2=s_y^2$

There are many examples similar to Normal distribution, like uniform and exponential family. And we will know that sufficient statistics for an exponential family is a minimal sufficient statistic.

1.4 Ancillary Statistics

Ancillary contains no information about θ. A statistic S(X) whose distribution does not depend on the parameter θ is called an ancillary statistic.

Illustrate ancillary statistics' role in a statistic.

Assume in a linear regression

$$Y = x_1 + x_2 - x_3 + \epsilon$$

$$\sigma = \begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{bmatrix}$$

so $cov(x_3, Y) = 0$. Predictions alone may be useless, but useful when combined with other variables

1.5 REMARK

Minimal Sufficient and Ancillary statistic Is a minimal sufficient statistic unrelated to an ancillary statistic?

Suppose $X_1, \ldots, X_n$ are i.i.d $U(\theta, \theta+1)$ random variables Then $\left(X_{(1)}, X_{(n)}\right)$ is a minimal sufficient statistic and the range statistic $R = X_{(n)} - X_{(1)}$ is an ancillary statistic

1.6 Complete Statistic

Firstly, we need to introduce a new definition of the Complete distribution family, and we will explain that the distribution of complete statistics belongs to the complete distribution family.

Definition 1.2 (Complete distribution family)**.** Assume $X \sim \mathcal{F} = \{f(x,\theta), \theta \in \Theta\}$. If for any measurable function $h(x)$, $\varphi(\theta) = E_\theta[h(X)] = \int h(x)dF_\theta(x) = 0, \forall\theta \in \Theta$, then we have $h(x) = 0, a.e.$. We define the distribution family is complete.

[2]

Example 1.2. Consider that Gamma family distribution $\mathcal{F} = \{\Gamma(\lambda, \nu)\}$ is complete.

We have $\varphi(\lambda, \nu) = \int_0^\infty h(x) \frac{\lambda^\nu}{\Gamma(\nu)} e^{-\lambda x} x^{\nu-1} dx = 0 \quad \forall \lambda, \nu$

Definition 1.3 (Fourier Transformation and Laplace Transformation)**.**

$$f(x) \to \int_{-\infty}^{\infty} e^{itx} f(x) d\mu(x)$$
$$f(x) \to \int_{-\infty}^{\infty} e^{-sx} f(x) d\mu(x)$$

This transformation is unique means that if this transformation's value is 0 always, then the corresponding left is 0 always. So we can use Laplace transformation here. Then we have $h(x) = 0, a.e.$. So we can say Gamma family distribution is complete.

Other complete distribution families, such as uniform distribution family have the same property.

Similarly, we can prove normal distribution is incomplete, when $h(x) = x$, the expectation of normal distribution is zero conflicts with the definition. And $T(X) = x^2$ where x is from normal distribution belongs to Chi-square distribution that is a complete distribution.

Definition 1.4 (Complete Statistic)**.** Let $f(t \mid \theta)$ be a family of pdfs or pmfs for a statistic $T(\mathrm{X})$. The family of a probability distributions is called complete if $E_\theta g(T) = 0$ for all θ implies $\mathrm{P}(g(T) = 0) = 1$ for all θ Equivalently $T(X)$ is called a complete statistic.

Complete Statistic is independent of any ancillary statistic. Now complete Statistic's definition is complicated. Don't forget the most crucial purpose of introducing this concept is to illustrate complete Statistic is independent on any ancillary statistic.

Theorem 1.4. *If a minimal sufficient statistic exists, then any complete statistic is also a minimal sufficient statistic.*

Theorem 1.5 (Basu's Theorem)**.** *If $T(x)$ is complete and minimal sufficient and statistics, T(x) is independent of every ancillary statistic.*

Proof. the proof only for discrete distributions. Let $S(X)$ be any ancillary statistic. Then $P(S(X))$ does not depend on θ since $S(X)$ is ancillary. Also, the conditional probability,

$$P(S(X) = s | T(X) = t) = P(X \in \{x : S(x) = s\} | T(X) = t)$$

does not depend on θ because $T(X)$ is a sufficient statistic.

Thus, to show $S(X)$ and $T(x)$ are independent, it suffices to show that

$$P(S(X) = s | T(x) = t) = P(S(X) = s)$$

for all possible values $t \in \mathcal{T}$. Now,

$$P(S(X) = s) = \sum_{t \in \mathcal{T}} P(S(X) = s | T(X) = t) P_\theta(T(X) = t)$$

Furthermore, since $\sum_{t \in \mathcal{T}} P_\theta(T(\mathbf{X}) = t) = 1$, we can write

$$P(S(\mathbf{X}) = s) = \sum_{t \in \mathcal{T}} P(S(\mathbf{X}) = s) P_\theta(T(\mathbf{X}) = t)$$

Therefore, if we define the statistic

$$g(t) = P(S(\mathbf{X}) = s \mid T(\mathbf{X}) = t) - P(S(\mathbf{X}) = s)$$

the above two equations show that

$$\mathrm{E}_\theta g(T) = \sum_{t \in \mathcal{T}} g(t) P_\theta(T(\mathbf{X}) = t) = 0 \quad \text{for all } \theta$$

since $T(\mathbf{X})$ is a complete statistic, this implies that $g(t) = 0$ for all possible values $t \in \mathcal{T}$. Hence the definition of complete statistic is verified. □

2 Chatper 7 Point Estimation

2.1 Estimator and Estimate

An estimator is a function of the sample, while an estimate is the realized value of an estimator that is obtained when a sample is actually taken.

2.2 Likelihood function and joint distribution density function

$$L(\theta|x) = \Pi f(x|\theta)$$

$$L\left(\theta_1, \theta_2 \mid \mathrm{x}\right) = \sum_{i=1}^{n} \log f\left(x_i \mid \theta_1, \theta_2\right)$$

2.3 Beset Unbiased Estimator

$$E(W - \theta)^2 = \mathrm{Var}(W) + \{E(W) - \theta\}^2 = \mathrm{Var}(W) + \{\mathrm{bias}(W)\}^2$$

Because we can't find uniformly optimal estimator controlling bias and variance, but we know the bias's error is smaller than that of variance in terms of magnitude. So we can find best estimator from unbiased estimator sets, which is named **UMVUE**.

2.4 Cramer-Rao Inequality

Theorem 2.1. *Let $X_1, X_2, \ldots$ be a sample with pdf $f(x|\theta)$, and let $W(x) = W(X_1, \ldots, X_n)$ be any estimator satisfying*

$$\frac{d}{d\theta} E_\theta W(X) = \int_{\mathcal{X}} \frac{\partial}{\partial\theta} \{W(x) f(x|\theta)\} \, dx \tag{1}$$

and $Var_\theta \{W(x)\} < \infty$. Then

$$Var_\theta \{W(x)\} \geq \frac{\left\{\frac{d}{d\theta} E_\theta W(x)\right\}^2}{E\left\{\frac{\partial}{\partial\theta} log f(x|\theta)\right\}^2} = \frac{\{\tau'(\theta)\}^2}{I(\theta)}$$

Proof. Subtitute w=1 into formula 1, we can get

$$0 = \frac{d}{d\theta} E_\theta(1) = \int_{\mathcal{X}} \frac{\partial}{\partial\theta} \{f(x \mid \theta)\} dx = E\left\{\frac{\partial}{\partial\theta} \log f(x \mid \theta)\right\}$$

Then, we have

$$E\left\{\frac{\partial}{\partial\theta} \log f(x \mid \theta)\right\}^2 = \mathrm{Var}\left\{\frac{\partial}{\partial\theta} \log f(x \mid \theta)\right\}$$

To apply the Cauchy-Schwartz inequality, it suffices to show that

$$Cov\left\{W(x), \frac{\partial}{\partial\theta} \log f(x|\theta)\right\} = E\left\{W(x) \frac{\partial}{\partial\theta} log f(x|\theta)\right\} = \tau'(\theta)$$

□

2.5 Score function

Definition 2.1 (score function)**.**

$$\frac{\partial}{\partial\theta} \log f(x|\theta)$$

An important property. If $f(x|\theta)$ satisfies

$$\frac{d}{d\theta} E_\theta \left\{\frac{\partial}{\partial\theta} \log f(X \mid \theta)\right\} = \int \frac{\partial}{\partial\theta} \left[\left\{\frac{\partial}{\partial\theta} \log f(x \mid \theta)\right\} f(x \mid \theta)\right] dx$$

then

$$E_\theta \left[\left\{\frac{\partial}{\partial\theta} \log f(X \mid \theta)\right\}^2\right] = -E_\theta \left[\left\{\frac{\partial^2}{\partial\theta^2} \log f(X \mid \theta)\right\}\right] \tag{2}$$

2.6 MLE is a good estimator?

The conclusion is that MLE is close to the C-R lower bound always. Now we will show that MLE is a good estimator.

Proof. Assume $\hat{\theta} \xrightarrow{P} \theta$, and we have

$$g(x|\theta) = \frac{\partial}{\partial\theta} \log f(x|\theta) = 0$$

Replace the expectation with sample mean.

$$\Rightarrow \frac{1}{n} \sum \left(g(x_i|\hat{\theta}) - g(x_i|\theta)\right) + \frac{1}{n} \sum g(x_i|\theta) = 0$$

The first term can be replaced by median theorem, then we have

$$\frac{1}{n}\sum g'(x_i|\tilde{\theta})(\hat{\theta}-\theta) = -\frac{1}{n}\sum g(x_i|\theta)$$
$$\Rightarrow Eg'(x|\theta)(\hat{\theta}-\theta) = -\frac{1}{n}\sum g(x_i|\theta) \to N(0, \frac{Var(g(x_i|\theta))}{n})$$

by $Eg(x) = 0$, we have

$$\hat{\theta}-\theta \to N(0, \frac{Var(g(x|\theta))}{n\left[E(g'(x|\theta))\right]^2})$$

by formula2, we have

$$\to N\left(0, \frac{1}{nVar(\frac{\partial log f(x|\theta)}{\partial\theta})}\right)$$

where

$$Var(\frac{\partial log f(x|\theta)}{\partial\theta}) = Var(\sum_i \frac{\partial log f(x_i|\theta)}{\partial\theta}) = \sum Var(\frac{\partial log f(x_i|\theta)}{\partial\theta})$$

So MLE is a good estimator. □

2.7 Rao-Blackwell Theorem

Theorem 2.2. *Let W be any unbiased estimator of $\tau(\theta)$, and let T be a sufficient statistic for θ. Define $\phi(T) = E(W|T)$. Then $\phi(T)$ is a uniformaly better unbiased estimator of $\tau(\theta)$*

Proof.

$$Var(W) = Var(E(W|T)) + E(Var(W|T)) \geq Var(\phi(T))$$

□

3 Chatper 8 Hypothesis Testing

3.1 Two types of errors

Though the probability of making the type-I error is very low, the hypothesis's result may be untrue when the sample is too unbalanced.

3.2 Likelihood Ratio Tests

The likelihood ratio test statistic for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_0^c$ is

$$\lambda(x) = \frac{\sup_{\Theta_0} L(\theta|x)}{\sup_{\Theta} L(\theta|x)}$$

Then the rejection region is $\lambda(x) \leq c$.

Example 3.1. Calculate the rejection region by

$$H_0 : \theta \leq \theta_0 \quad v.s. \quad H_1 : \theta > \theta_0$$

$$\begin{aligned} L(\theta|X) &= (\frac{1}{\sqrt{2\pi}\sigma}^{n})e^{-\frac{\sum_i (x_i-\theta)^2}{2\sigma^2}} \\ logL &= nlog(\frac{1}{\sqrt{2\pi}\sigma}) - \frac{\sum_i (x_i-\theta)^2}{2\sigma^2} \\ \frac{\partial logL}{\partial \theta} &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\theta) \end{aligned} \tag{3}$$

If $\bar{x} \leq \theta_0$ then $\lambda(x) = 1$, but can't reject.

If $\bar{x} > \theta_0$, then $\bar{x} > \theta_0 > \theta$, so maximum of numerator is $L(\theta_0|x)$

so

$$\begin{aligned} \lambda = \frac{L(\theta_0|x)}{L(\bar{x}|x)} &= e^{-\frac{n(\bar{x}-\theta_0)^2}{2\sigma^2}} \leq c \\ \bar{x} &\geq \theta_0 + \sqrt{\frac{-2\sigma^2 logc}{n}} \end{aligned} \tag{4}$$

3.3 Power function

Definition 3.1. The power function of a hypothesis test with rejection region R is the function of θ defined by $\beta(\theta) = P_\theta(x \in R)$

The ideal power function is 0 for all $\theta \in \Theta_0$ and 1 for all $\theta \in \Theta_0^c$

Example 3.2. Let $X_1, \ldots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ population, σ^2 known.

An LRT of $H_0 : \theta \leq \theta_0$versus $H_1 : \theta > \theta_0$ is a test that rejects H_0 if $(\bar{X} - \theta_0)/(\sigma/\sqrt{n}) > c$ The power function is

$$\beta(\theta) = P_\theta(\frac{\bar{X}-\theta_0}{\sigma/\sqrt{n}} > c) = P_\theta(Z > c + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}})$$

where Z is a standard normal random variable.

We will $\beta(\theta)$ is a non-decreasing function here. When H_0 is right, the maximum of $\beta(\theta)$ is $\beta(\theta_0) = P(Z > c)$, the ideal circumstance is 0 by controlling c.

Next we hope the minimum of $\beta(\theta)$ is 1 when H_1 is right. Relax the H_1, assume $H_1 : \theta \geq \theta_0 + \tau \Rightarrow \beta(\theta) \geq \beta(\theta_0 + \tau) = P(z > c + \frac{-\tau\sqrt{n}}{\sigma}) \overset{n\to\infty}{\to} 1$

So, if n is large enough, and τ is small enough, the power function will be ideal because the β value interval is very tiny between [0,1], which is just τ length.

Definition 3.2 (Unbiased Test)**.** A test with power function $\beta(\theta)$ is unbiased if $\beta(\theta') \geq \beta(\theta'')$ for every $\theta' \in \Theta_0^c$ and $\theta'' \in \Theta_0$.

Equivalently, if the power function $\beta(\theta) = P_\theta(x \in R)$ satisfies:

$$\beta_\theta \leq \alpha, \theta \in \Theta_0; \beta_\theta > \alpha, \theta \in \Theta_0^c$$

Definition 3.3 (Test function)**.** $\phi(x) = \begin{cases} 1 & x \in R \\ 0 & x \notin R \end{cases}$ where R is a rejection region.

Definition 3.4 (UMP level α test)**.** Let C be class of all level α tests for testing. A test in class C, with power function $\beta(\theta)$ is a uniformly most powerful(UMP) class C test if $\beta(\theta) \geq \beta'(\theta)$ for every $\theta \in \Theta_0^c$, and every $\beta'(\theta)$ that is a power function of a test in class C.

Turn the definition into a graph to explain that the power function is highest at H_1, but no restriction in H_0 except α.

3.4 Neyman-Pearson Theorem

Definition 3.5 (size α test)**.** For $0 \leq \alpha \leq 1$, a test with power function $\beta(\theta)$ is a size α test if $\sup_{\theta \in \Theta_0} \beta(\theta) = \alpha$

Definition 3.6 (level α test)**.** For $0 \leq \alpha \leq 1$, a test with power function $\beta(\theta)$ is a size α test if $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$

Theorem 3.1. *Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ where the pdf ol pmf corresponding to θ_i is $f(x \mid \theta_i), i = 0, 1$, using a test with rejection region R that satisfies*

$$\begin{aligned} x \in R \text{ if } f(x \mid \theta_1) > k f(x \mid \theta_0) \\ x \in R^c \text{ if } f(x \mid \theta_1) < k f(x \mid \theta_0) \end{aligned} \tag{5}$$

In fact, the two formulas are the equivalent.

For some $k \geq 0$

$$\alpha = \mathrm{P}_{\theta_0}(x \in R) \tag{6}$$

Then,

1. *Sufficiency: Any test that satisfies 5 and 6 is a UMP level α test.*
2. *Necessity: If there exists a test satisfying 5 and 6 with $k > 0$, then every UMP level α test is a size α(**remark: distinguish concepts here because UMP level α means the type I error is not compact**) and every UMP level α test satisfying 5 except perhaps on a set A satisfying $P_{\theta_0}(x \in A) = P_{\theta_1}(x \in A) = 0$*

Proof. Sufficiency: Because the test is a simple hypothesis test, so it's a size α test if it satisfies 6 (However, it's not a UMP).

Now we prove that it is a UMP. Define a test function, a function on the sample space that is 1 if $x \in R$ and 0 if $x \in R^c$, and let $\phi(x)$ be the test function of a test satisfying 5 and 6,

$$\phi(x) = \begin{cases} 1 & x \in R(f(x|\theta_1) > k f(x|\theta_0)) \\ 0 & x \in R^c(f(x|\theta_1) < k f(x|\theta_0)) \end{cases} \quad \phi'(x) = \begin{cases} 1 & x \in R' \\ 0 & x \in R'^c \end{cases} \tag{7}$$

$\phi'(x)$ is an arbitrary test function which is level α. by 5. Since $\phi = 1$ if $f(x|\theta_1) > k f(x|\theta_0)$ and $\phi = 0$ if $f(x|\theta_1) < k f(x|\theta_0)$. we have

$$0 \leq \int [\phi(x) - \phi'(x)][f(x|\theta_1) - k f(x|\theta_0)] dx = \int I_{\{\phi(x) - \phi'(x)\}}[f(x|\theta_1) - k f(x|\theta_0)] dx$$

recall the definition of β power function 3.1

$$= \beta(\theta_1) - \beta'(\theta_1) - k[\beta(\theta_0) - \beta'(\theta_0)] \tag{8}$$

Noting that $\phi'(x)$ is a level α test, but $\phi(x)$ is a size α test. So we have

$$\beta(\theta_0) - \beta'(\theta_0) = \alpha - \beta'(\theta_0) \geq 0$$

Combine above formula, we have

$$0 \leq \beta(\theta_1) - \beta'(\theta_1)$$

Necessity:

To prove necessity, let ϕ' now be the test function for any UMP level α test. Assume a test ϕ satisfy 5 and 6, is also a UMP level α test. So the goal of us is to prove ϕ' is a size α test.

Since ϕ and ϕ' are UMP level α test, then $\beta(\theta_1) = \beta'(\theta_1)$. This fact, 8, and k>0 imply

$$\alpha - \beta'(\theta_0) = \beta(\theta_0) - \beta'(\theta_0) \leq 0$$

Now, since ϕ' is a level α test, $\beta'(\theta_0) \leq \alpha \Rightarrow \beta'(\theta_0) = \alpha$. That is ϕ' is a size α test, and this is also implies that 8 is an equality in this case. But the non-negative integrate $(\phi(x) - \phi'(x))(f(x|\theta_1) - kf(x|\theta_0))$ will have a zero integral only if ϕ' satisfies 5, except perhaps on a set A with $\int_A f(x|\theta_i)dx = 0$. This implies that the last assertion in statement (b) is true.

□

recall a lemma

Lemma 3.2. *Assume $T = T(x)$ is a sufficient statistic, any test $\phi(x)$, there exists a test $\tilde{\phi}(T)$ is a function of T, and it has the same effect to $\phi(x)$*

Corollary. *Suppose $T(x)$ is a sufficient statistic for θ and $g\ (t \mid \theta_i)$ is the pdf or pmf of $T(x)$ corresponding to $\theta_i, i = 0, 1$. Then any test based on $T(x)$ with rejection region S (a subset of the sample space of T) is a UMP level α test if it satisfies*

$$\begin{cases} T(x) \in S & \text{if} \quad g\left(T(x) \mid \theta_1\right) > kg\left(T(x) \mid \theta_0\right), \text{ and} \\ T(x) \in S^c & \text{if} \quad g\left(T(x) \mid \theta_1\right) < kg\left(T(x) \mid \theta_0\right) \end{cases}$$

for some $k \geq 0$ and

$$\alpha = \mathrm{P}_{\theta_0}(g(T(\mathrm{x}) \in S)$$

Theorem 3.3 (Karlin-Rulbin Theorem)**.** *Consider testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$. Suppose that T is a sufficient statistic for θ and the family of pdfs or pmfs $\{g(t \mid \theta) : \theta \in \Theta\}$ of T has an nondecreasing* MLR. *Then for any t_0, the test that rejects H_0 if and only if $T > t_0$ is a UMP level α test, where $\alpha = \mathrm{P}_{\theta_0}(T > t_0)$*

Karlin-Rulbin theorem is the extension of the Neyman-Pearson Theorem, and the non-decreasing condition is sufficient condition but not necessity condition.

Proof. 1. extension of the null hypothesis to two points.

$$H_0 : \theta \in \left\{\theta_0, \theta_0'\right\} \quad vs \quad H_1 : \theta \in \{\theta_1\}$$

then we hope $\beta(\theta_0) \leq \alpha$, $\beta(\theta_0') \leq \alpha$ and $\beta(\theta_1) \geq \tilde{\beta}(\theta_1)$. The primal test have satisfied the first and third term. Now the key is to verify whether satisfy the second term(considering Monotone likelihood ratio(**MLR**)).

2. extension of the alternative hypothesis.

$$H_0 : \theta \in \{\theta_0\} \quad vs \quad H_1 : \theta \in \{\theta_1, \theta_2\}$$

then we hope $\beta(\theta_0) \leq \alpha$ and $\beta(\theta_1) \geq \tilde{\beta}(\theta_1)$, $\beta(\theta_2) \geq \tilde{\beta}(\theta_2)$ If $\beta(\theta_0) \leq \alpha$ and $\beta(\theta_1) \geq \tilde{\beta}(\theta_1)$ is right like a two points test, we have $\frac{f(x|\theta_1)}{f(x|\theta_0)} \geq K$. Similarly, we have $\frac{f(x|\theta_2)}{f(x|\theta_0)} \geq K'$. If the two inequalities indicate rejection regions are the same(or the rejection region is independent of parameters in alternative hypothesis), then the extension is rational(considering Monotone likelihood ratio(**MLR**)).

So when **MLR** is non-decreasing($H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$), the two extension holds. So non-decreasing **MLR** is a sufficient condition. □

Remark: level α test means α is not a compact, but size α means it's a compact.

3.5 Nonexistence of UMP test

Although most experimenters would choose to use a UMP level test if they knew of one, unfortunately, for many problems, there is no UMP level test.

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

The test that rejects H_0 if $\bar{X} < \frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0$ has the highest possible power at $\theta_1 < \theta_0$

The test that rejects H_0 if $\bar{X} > -\frac{\sigma z_\alpha}{\sqrt{n}} + \theta_0$ has the highest possible power at $\theta_2 > \theta_0$

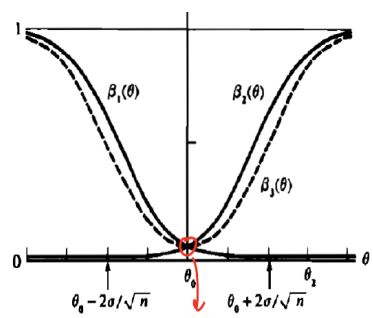

Figure 1: UMP unbiased level αtest

The natural idea is that why not choose the bell-shaped formed by the real segments of $\beta_1(\theta)$ and $\beta_2(\theta)$. The key here is that this situation is can not reachable by the lowest point would be lower than before, so it's not unbiased.

3.5.1 Why is the situation unattainable?

In other words, consider why the left section in power function of test2 is close to zero if right section is the real line of $\beta_2(\theta)$.

Firstly ,we need to know that $\beta_1(\theta)$ corresponding to test $H_0 : \theta \geq \theta_0$ v.s. $H_1 : \theta < \theta_0$, and $\beta_2(\theta)$ corresponding to test $H_0 : \theta \leq \theta_0$ v.s. $H_1 : \theta > \theta_0$. Test1's rejection region is $\left\{\bar{X} > \sigma z_\alpha/\sqrt{n} + \theta_0\right\}$. Test2's rejection region is $\left\{\bar{X} < \sigma z_\alpha/\sqrt{n} + \theta_0\right\}$

And test1 can also be written as

$$H_0 : \theta = \theta_0 \quad v.s. \quad H_1 : \theta < \theta_0$$

Consider one point θ_1 in H_1(by **MLR**)

$$H_0 : \theta = \theta_0 \quad v.s. \quad H_1 : \theta = \theta_1$$

where $\theta_1 < \theta_0$

By Neyman-Pearson Theorem first conclusion, the rejection region $\left\{\bar{X} > \sigma z_\alpha/\sqrt{n} + \theta_0\right\}$ is the UMP size α's rejection region.

By Neyman-Pearson Theorem second conclusion, any other level α test that has as high power as Test1 at θ_1 must have the same rejection region as Test1 except possibly for a set A satisfying $\int_A f(x|\theta_i)dx = 0$. Thus if a UMP level α test exists for this problem, it must be Test1, because no other test has a high a power as Test1 at θ_1. If the rejection region is guaranteed, then the power function of the right part is also determined.

Remark: Neyman-Pearson theorem guarantees that when the UMP size α is determined, then the rejection region is determined and the test is determined.

3.6 P-value

Definition 3.7. Let $W(X)$ be a test statistic such that large values of W give evidence that H_1 is true. For each sample point x, define

$$p(x) = \sup_{\theta \in \Theta_0} P_\theta(W(X) \geq W(x))$$

Then, p(x) is a valid p-value.

The $W(X)$ here is the cumulative distribution function, which is a uniform distribution.

A p-value $p(x)$ is a test statistic satisfying $0 < p(x) < 1$ for every sample point x. Small values of $p(x)$ give evidence that H_1 is true.

Note that if you are not a Bayesian school, the p-value is not the same as the null hypothesis's probability is right.

3.6.1 Regression's drawbacks

In regression, the null hypothesis is $x == 0$, which is a simple hypothesis. So when we consider $p < 0.05$ is the criterion in the test. It has two drawbacks:

1. $W(X)$ is a uniform distribution, and it maybe makes the type I error. This would happen 20 times on average.
2. why compare with 0.05, assume it is uniform distribution is too strong here. Maybe it will be an excellent β function here.

4 Chapter 9 Interval Estimation

4.1 Pivotal Quantities

Definition 4.1 (Pivotal Quantities)**.** A random variable $Q(x, \theta) = Q(X_1, \ldots, X_n, \theta)$ is a pivotal quantity if the distribution of $Q(x, \theta)$ is independent of all parameters. That is, if $x \sim F(x|\theta)$, then $Q(x, \theta)$ has the same distribution for all values of θ

Remark: Q is a function of θ, so it is not a statistic.

4.1.1 Location-scale Pivots

Location: $f(X - \mu)$ Pivotal quantity: $\bar{X} - \mu$
Scale: $\frac{1}{\sigma} f(\frac{X}{\sigma})$ Pivotal quantity:$\frac{\bar{X}}{\sigma}$
Location-scale: $\frac{1}{\sigma} f(\frac{x-\mu}{\sigma})$ Pivotal quantity:$\frac{\bar{X}-\mu}{S}$

4.1.2 Gamma Pivot

Suppose that $X_1, \ldots, X_n$ are i.i.d exponential (λ). Then $T = \sum X_i$ is a sufficient statistic for λ and $T \sim \Gamma(n, \lambda)$ Thus, if $Q(T, \lambda) = 2T/\lambda \sim \Gamma(n, 2) = \chi^2_{2n}$ We choose constant a and b to satisfy $\Pr\left(a \leq \chi^2_{2n} \leq b\right) = 1 - \alpha$

Inverting the set $A(\lambda) = \left\{t : a \leq \frac{2t}{\lambda} \leq b\right\}$ gives $C(t) = \left\{\lambda : \frac{2t}{b} \leq \lambda \leq \frac{2t}{a}\right\}$, which is a $1 - \alpha$ confidence interval.

4.1.3 A general method to find pivot

In general, suppose the pdf of a statistic T, $f(t|\theta)$ can be expressed in the form

$$f(t \mid \theta) = g(Q(t, \theta)) \left| \frac{\partial}{\partial t} Q(t, \theta) \right| \tag{9}$$

for some function g and some monotone function Q.(monotone in t for each θ)

Utilize cumulative density function to transform it.

4.2 Evaluating Interval Estimators

Longer is confidence interval, the smaller probability of making the first type of error. Shorter is confidence interval, the smaller probability of making the second type of error.

Make the confidence interval is small enough under controlling type I error.

4.3 unimodal pdf

Let f (x) be a **unimodal pdf**, and optimal length is correlerated with b-a. If the interval [a, b] satisfies

1. $$\int_a^b f(x)dx = 1 - \alpha \tag{10}$$

2. $$f(a) = f(b) > 0 \tag{11}$$

3. $$a \leq x^* \leq b \tag{12}$$

, where x^* is a mode of $f(x)$

then [a,b] is the shortest among all intervals that satisfy (1)

4.4 optimal length is not correlerated b-a

Suppose $X \sim \Gamma(k, \beta)$. The quantity $Y = X/\beta$ is a pivot with $Y \sim \Gamma(k, 1)$, so we can get a confidence interval by finding constants a and b to satisfy

$$\Pr(a \leq Y \leq b) = 1 - \alpha$$

The interval of β is of the form

$$\left\{\beta : \frac{X}{b} \leq \beta \leq \frac{X}{a}\right\}$$

so the length of the interval is $\left(\frac{1}{a} - \frac{1}{b}\right) X$; that is, it is proportional to $(1/a) - (1/b)$ and not to $b - a$.

We must solve the following constrained minimization problem

$$\begin{aligned} &\min \left(\tfrac{1}{a} - \tfrac{1}{b}\right) \\ \text{subject to} \quad &\textstyle\int^b f(y)dy = 1 - \alpha \end{aligned}$$

Differentiating the first equation with respect to a and setting it to 0 yield the identity $db/da = b^2/a^2$. Substituding this in the derivative of the second equation which must equal to 0, gives $f(b)b^2 = f(a)a^2$.

References

[1] G. Casella and R. L. Berger, *Statistical inference*, vol. 2, Duxbury Pacific Grove, CA, 2002.

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